

Scan In and License Plate a Carton SOP

One episode scans in and license plates one carton. The carton is already sitting on the work spot when recording starts. Its barcode is scanned to a good read, a license plate label is applied square on its top face, and the carton is staged to the lane spot. The work spot is bare and the carton stands labeled in the lane when the episode ends.

The order never changes: scan, label, stage. A label is never applied to a carton that has not read, and the carton is never staged before its label is on.

The left arm supports and the right arm works. The left gripper anchors the carton for the label press and pushes the carton toward the lane in Step 3. The right gripper works the scanner and the label, then guides the carton during that push. Neither arm crosses the middle of the desk to take the other's zones.

The carton is not moved at all until Step 3. It stays flat on the desk while both grippers slide it to the lane in the same facing it started in. It is never lifted, turned, or tilted. No gripper ever closes on the top face of the carton or on the label.

The scanner never touches the carton, the label is only ever handled by its grab tab, a label that has touched the carton is never peeled up and stuck down again, and no gripper opens anywhere but over the spot the thing in it is going to.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole desk: the work spot at the middle, the lane spot on the right, and the peel board and the scanner rest at the front right.
3. The top face of the carton is in frame from above, with the patch outline and all four edges of a label visible on it.
4. The near face of the carton is in frame, so the barcode and the scanner head are both visible during a read.
5. The scanner's indicator lamp is in frame whenever the scanner is in the gripper.
6. Both arms are at home with grippers open.
7. The desk is clear of anything but the zones listed below.

8. The right gripper reaches the carton's top face, near face, right wall, scanner rest, peel board, and lane spot without the arm leaning out.
9. The left gripper reaches the carton's left wall and the whole push path from the work spot to the lane spot without reaching a joint limit.
10. The lane spot sits immediately right of the work spot at the same depth. The straight path between them is bare, and both grippers can travel beside the carton without touching each other.

Materials checklist

1. One carton already resting flat and square on the **work spot**, closed and taped, with its barcode face toward the front edge. Nothing else stands on the desk except the zones below.
2. The carton has a flat, clean base that slides across the desk under a steady push without catching, tipping, or needing to be lifted.
3. The carton carries a printed **barcode label** low on its near face, clear of the top edge, showing a barcode that reads on the station scanner.
4. The carton has a printed **patch** outline in the middle of its top face, a little larger than a label, with its edges in line with the carton's edges. The patch is bare.
5. Two **license plate labels** on the peel board, sticky side down, in a row with the one nearest the work spot used first and the second held in reserve for a scrapped label. Each is a square carrying a printed license plate code, each peels off in one piece, and each has a stiff folded back **grab tab** along one edge.
6. One **scanner** lying on the scanner rest, handle toward the front edge and head away. It is switched on and set to read continuously, so no trigger is pulled, and a good read shows one green flash of the indicator lamp with one beep.
7. The **lane spot** is bare, with its printed outline readable.

Workspace layout

- **Work spot:** a marked square at the middle of the desk, centered between the two arms. The carton starts here and stays here until Step 3.
- **Lane spot:** a marked square immediately right of the work spot at the same depth. The finished carton is slid here along one straight, bare path and nothing else is ever set on it.
- **Peel board:** at the front right, between the work spot and the scanner rest, holding the two labels. Its surface releases a label cleanly when the grab tab is lifted.
- **Scanner rest:** the front right corner, outboard of the peel board. The scanner lies here before and after the read.

The left arm's zone is the carton's left wall. The right arm's zones are the carton's top face, near face, and right wall, the peel board, the scanner rest, and the lane spot. Step 3 is the only

shared travel: both grippers move right with the carton while it stays on the desk.

Vocabulary

- **Arm assignment:** the left arm anchors and supplies the push in Step 3. The right arm scans, labels, and guides the carton during the push. Neither arm takes the other's job. The two work together during the label press and the final carton slide.
- **Barcode label:** the printed label on the carton's near face. It carries the barcode that is scanned. It is never covered, moved, or touched by a gripper.
- **License plate label:** one square label from the peel board, called the **label** below. It is the only thing applied to the carton.
- **Patch:** the printed outline on the carton's top face that the label is laid inside.
- **Square:** the label lying inside the patch with each of its edges in line with the matching edge of the patch, no corner overhanging the outline, and no corner lifted.
- **Grab tab:** the folded back, non sticky edge of a label. It is the only part of a label a gripper ever touches.
- **Hangs flat:** the label dangling free and flat from the lifted grab tab, not curled, folded, or stuck to itself. A label that is not hanging flat is scrapped, not straightened.
- **Apply:** the right gripper lowering a hanging label straight down into the patch and releasing the grab tab once the label is lying on the top face.
- **Press:** the closed right gripper set down flat on the applied label, held still for 2 seconds, and lifted straight off. A press never slides and never rubs.
- **Scanner:** the tool on the scanner rest. Its **handle** has a flat grip section and its **head** carries the read window and the indicator lamp. A gripper closes on the handle and never on the head.
- **Good read:** one green flash of the indicator lamp with one beep, taken while the head is held still and square to the barcode. Nothing else counts as a read.
- **Anchor:** the closed left gripper pressed against the outside of the carton's left wall, level with the middle of the wall, holding the carton still while the right gripper presses the label. An anchor never travels, never sits on the top face, and lifts clear only after the right gripper is off the carton.
- **Guided slide: Unvalidated at the station.** Both grippers closed before contact. The left gripper presses against the middle of the carton's left wall and supplies a slow push toward the lane. The right gripper rests against the middle of the right wall and travels with the carton as a guide, preventing it from turning. The base stays flat on the desk for the whole move. Neither gripper closes around the carton.
- **Lift:** taking the scanner or a label straight down onto its grasp, closing, and raising it straight up until there is daylight under it, before anything moves sideways.

- **Set down:** carrying the thing level to its spot, lowering it straight down until it is resting flat, and releasing once it is resting.
- **Facing:** the carton's barcode face pointing at the front edge. The carton keeps its facing from the work spot to the lane spot.
- **Settled:** the carton stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, or falling over.
- **Release point:** over the patch or the scanner rest, once the thing in the gripper is already resting. A gripper opens nowhere else. The carton is pushed with already-closed grippers and is never held or released.

Steps

Steps 1 to 3 are one carton. The carton is already on the work spot, so the episode starts with the scanner, not with a pick.

Step 1: Scan the barcode

Goal: a good read on the carton's barcode, with the scanner back on its rest.

- The **right gripper** comes down onto the flat of the **handle** on the scanner rest, closes across it, and lifts the scanner straight up.
- Bring the **head** square to the **barcode** on the carton's near face, close to it but not touching it, and hold it still until the lamp flashes green and the scanner beeps once.
- Lift the scanner straight back, carry it to the **scanner rest**, lower it flat with the handle toward the front edge, and release.
- The right gripper lifts clear.

The head is held still and square to the label. If the lamp does not flash, draw the head back, bring it square to the barcode again, and hold for a second read. Never touch the carton with the scanner, and never move the carton to bring its barcode to the head.

Check: the lamp flashed green and the scanner beeped once while the head was on the barcode, the scanner lies flat on its rest with the handle toward the front edge, and the carton sits exactly where it started with its facing unchanged. If three attempts give no read, the barcode is defective: stop, log it as a system issue, and discard the episode.

Step 2: Apply the license plate label square

Goal: the label lying square inside the patch on the carton's top face, pressed flat.

- The **left gripper**, closed, **anchors** the carton on its **left wall** and holds it there for the whole step.
- The **right gripper** pinches the **grab tab** of the label nearest the work spot, lifts it straight up off the peel board until it hangs flat and clear, and carries it to the carton without turning it.

- Lower the label straight down into the **patch**, with its edges in line with the patch's edges, until it is lying on the top face. Release the grab tab.
- The **right gripper**, closed, presses down flat on the label, holds for 2 seconds, and lifts straight off.
- Both grippers lift clear.

Line the label up over the patch before you lower it. Adhesive sticks once, so a label is placed by lowering it, never by sliding it into line after it has touched the carton.

Check: the label lies inside the patch, square, with no corner overhanging the outline, no corner lifted, no fold, and no bubble; the barcode label on the near face is uncovered; and the carton is still square on the work spot with its facing unchanged. If the label landed off the patch, crooked, or folded on itself, peel it off, scrap it, and apply the reserve label from the peel board. A label is never peeled up and stuck down again.

Step 3: Slide the carton to the lane

Goal: the carton resting on the lane spot, facing the front edge, with its label up and readable.

- Close both grippers before either one touches the carton.
- The **left gripper** presses against the middle of the carton's **left wall**. The **right gripper** rests against the middle of its **right wall**, clear of the top face and the label.
- Make one slow **guided slide** straight right until the carton is fully inside the **lane spot**. The left gripper supplies the push; the right gripper travels with the right wall and keeps the carton square.
- Stop once the carton is centered in the lane spot, then lift both grippers straight clear.

This is the first and only time the carton is moved. Its base stays on the desk, its facing does not change, and neither gripper pinches, lifts, or closes across the carton.

Check: the carton rests flat and square inside the lane spot, settled, its barcode face toward the front edge, its label up, square, and readable from above, and the work spot is bare. If the carton stops short or starts to turn, slide it back to the work spot with the right gripper supplying the reverse push and the left gripper guiding, then repeat the guided slide.

Step 4: End the episode

- Confirm the carton stands on the lane spot facing the front edge with its label square in the patch, the work spot is bare, and the scanner lies on its rest with the handle toward the front edge.
- Return both arms home, clear of the carton and the lane spot, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the carton off the lane spot, peel the label off its top face, and clear the scrapped labels off the desk. Confirm the patch outline is still readable.
2. Reload the peel board with two fresh labels, sticky side down, the first nearest the work spot, and check that both grab tabs are still folded back and do not stick.
3. Set a carton flat and square on the work spot with its barcode face toward the front edge and its patch bare.
4. Check the scanner: switched on, set to read continuously, and giving one green flash and one beep on a good barcode. Lay it flat on the scanner rest, handle toward the front edge.
5. Replace anything damaged: a carton that will not sit square or is crushed, a barcode label that is torn, smudged, or unreadable, a label that has lost its tack, or a peel board that no longer releases a label in one piece.
6. Wipe the desk and confirm the work spot, the lane spot, the peel board, and the scanner rest have not shifted and that every printed outline is readable.
7. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the label is applied before the carton has read, or the carton is moved to the lane spot before Step 3.
- **SOP rule broken:** Steps 1 to 3, the order never changes: scan, label, stage.
- **Coaching note:** finish the step you are in before you start the next one.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the scanner or the label, or the right gripper supplies the forward push while the left guides.

- **SOP rule broken:** Steps 1 to 3, the left arm anchors and supplies the forward push, and the right arm scans, labels, and guides the carton during that push.
- **Coaching note:** left holds and pushes; right scans, labels, and guides.

Violation: Carton moved before Step 3

- **Visible cue:** the carton slides, turns, tips, or is lifted off the work spot during the scan or the label, including being moved to bring its barcode to the scanner head.
- **SOP rule broken:** Steps 1 and 2, the carton sits still on the work spot and is not moved at all until Step 3.
- **Coaching note:** the tool moves, the carton does not.

Violation: Carton gripped on the top face or on the label

- **Visible cue:** a gripper closes on, presses on, or drags across the carton's top face, its patch, or the applied label during the anchor or guided slide.
- **SOP rule broken:** Steps 2 and 3, grippers contact the outside side walls only and never the top face or the label.
- **Coaching note:** closed grippers against side walls only. The top face belongs to the label.

Violation: Carton turned or tilted

- **Visible cue:** the carton rotates, tips, or leaves the desk during the guided slide, or finishes with its barcode face pointing anywhere but toward the front edge.
- **SOP rule broken:** Step 3, the carton base stays flat on the desk and its facing remains unchanged from the work spot to the lane spot.
- **Coaching note:** push slowly from the left wall and keep the right-wall guide travelling with it.

Violation: Carton lifted or pinched during the slide

- **Visible cue:** either gripper opens around an edge or wall, closes on the carton, or raises any part of the carton clear of the desk during Step 3.
- **SOP rule broken:** Step 3, both grippers are already closed before contact and slide the carton with its base flat on the desk; neither gripper pinches or lifts it.
- **Coaching note:** use the closed grippers as push faces. The desk supports the carton.

Violation: Scanner mishandled

- **Visible cue:** a gripper closes on the head or across the scanner at an angle instead of on the flat of the handle; the head touches the carton, the barcode, or the desk; or the scanner is left on the carton, in the lane, or still in the gripper when the label is taken, or dropped onto its rest rather than lowered.
- **SOP rule broken:** Step 1, the flat of the handle is the only part gripped, the head touches nothing, and the scanner is back flat on its rest before the label is taken.

- **Coaching note:** handle only, hover the head, then put the tool back.

Violation: Moved on without a good read

- **Visible cue:** the scanner is returned and the episode goes on to the label with no green flash and no beep, or waves the head past the barcode without holding it still.
- **SOP rule broken:** Step 1, only one green flash with one beep counts as a read, and Step 2 is never started on a carton that has not read.
- **Coaching note:** hold it still and wait for the flash. No flash, no label.

Violation: Label taken by the wrong part

- **Visible cue:** a gripper pinches the label anywhere but its grab tab, touches the sticky face, or peels it off the board by dragging it rather than lifting the tab.
- **SOP rule broken:** Step 2, the grab tab is the only part of a label a gripper touches.
- **Coaching note:** tab only. A label that has touched the gripper is already ruined.

Violation: Label applied curled or folded

- **Visible cue:** the label is curled, twisted, or stuck to itself as it is carried and is applied anyway, or a gripper tries to unstick a folded label in the air instead of scrapping it.
- **SOP rule broken:** Step 2, the label is lifted until it hangs flat and clear, and one that is not hanging flat is scrapped, not straightened.
- **Coaching note:** look at it hanging before you bring it over. A folded label is scrap.

Violation: Label not square

- **Visible cue:** the label lands outside the patch, overhangs the outline at a corner, sits at an angle to the patch edges, hangs over the edge of the carton, or covers the barcode label.
- **SOP rule broken:** Step 2, the label lies inside the patch with its edges in line with the patch's edges and no corner overhanging.
- **Coaching note:** line the label up over the patch before you lower it. The landing decides it.

Violation: Label peeled up and stuck down again

- **Visible cue:** the applied label is lifted off and pressed back down, slid into line after contact, or repositioned by peeling a corner.
- **SOP rule broken:** Step 2, a badly applied label is peeled off, scrapped, and replaced with the reserve label.
- **Coaching note:** adhesive sticks once. Scrap it and take the reserve.

Violation: Press skipped or rubbed

- **Visible cue:** the episode moves on from an applied label with no press, presses for less than the hold, dabs at it in several places, or slides the gripper across the label instead of holding it still.

- **SOP rule broken:** Step 2, one flat press held for 2 seconds and lifted straight off, never slid and never rubbed.
- **Coaching note:** down flat, count two, straight up.

Violation: Pressed without an anchor

- **Visible cue:** the label is applied and pressed with the left gripper off the carton, or the anchor is taken only after the press has started, or it lets go before the right gripper is clear.
- **SOP rule broken:** Step 2, the left gripper anchors the left wall first and holds it for the whole step.
- **Coaching note:** anchor first, let go last.

Violation: Anchor in the wrong place

- **Visible cue:** the anchoring gripper presses on the top face, on the label, on the barcode face, or on a wall the right gripper is working, or it slides along the wall during the press.
- **SOP rule broken:** Step 2, the anchor sits still on the outside of the carton's left wall and never on the top face.
- **Coaching note:** flat on the side wall, and hold it still.

Violation: Item dropped instead of lowered

- **Visible cue:** a gripper opens above the desk and the scanner falls the rest of the way, or it is set down short of its rest and slid into place.
- **SOP rule broken:** Step 1, release the scanner only once it is resting.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the peel board, over the bare desk, or over open space, and what it held lands anywhere but on the spot it was going to.
- **SOP rule broken:** Steps 1 to 3, a gripper opens over the patch or the scanner rest and nowhere else; the carton is moved with already-closed grippers and is never released.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a label is slid after contact; or the carton is corrected with small pats instead of the complete reverse-and-repeat slide prescribed in Step 3.
- **SOP rule broken:** Steps 2 and 3, scrap a misplaced label and return a misplaced carton to the work spot before repeating the guided slide.
- **Coaching note:** use the prescribed retry from its starting state, not small corrective shoves.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or seats its hold again on a label or the scanner without setting it down first. Side-wall contact during the guided slide is not a grasp or regrasp.
- **SOP rule broken:** Steps 1 and 2, set a held item down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** the carton, the label, or the scanner falls onto the desk or the floor; the peel board or the scanner rest is struck and shifts; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 3, keep a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the carton still on the work spot, the label still on the peel board, the scanner off its rest, an arm away from home, or recording stops before both arms are home.
- **SOP rule broken:** Step 4, confirm the carton, the bare work spot, and the scanner; return both arms home; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Scanner fault:** the lamp does not light, the scanner loses power, it beeps with no code in front of it, or it will not read a clean barcode held still and square after three attempts. Replace the scanner before the next episode.
- **Defective barcode label:** torn, smudged, peeling off the carton, or printed too faint to read.
- **Defective label:** lost tack, torn on lifting, cut to the wrong size for the patch, or a grab tab that has come unfolded. Replace the label before the next episode.
- **Peel board will not release** a label in one piece.
- **Defective carton:** crushed, will not sit square, catches or tips during a correct guided slide, or its top face patch is missing, misprinted, or already carrying a label.
- **The work spot, the lane spot, the peel board, or the scanner rest shifts out of place** during the episode.

Annotation subtasks (from SOP)

1. Take the scanner from its rest
2. Scan the carton barcode to a good read
3. Return the scanner to its rest
4. Lift one license plate label from the peel board
5. Apply the label into the patch and press it
6. Slide the carton to the lane spot with both grippers
7. Confirm the lane and end the episode